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Air pollution and Parkinson's disease: A prospective cohort study with sex-stratified analysis in the UK biobank

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ABSTRACT

Air pollution has been suggested as a potential environmental risk factor for Parkinson's disease (PD), but findings remain inconsistent, and sex-specific effects are understudied. This study examined associations between exposure to nitrogen dioxide (NO_2) and particulate matter ≤ 10 µm (PM_{10}) and incident PD, using data from the UK Biobank. Annual levels of NO_2 (2005–2007) and PM_{10} (2007) were estimated based on residential addresses. Logistic regression models assessed the associations between air pollution exposure and PD onset, adjusting for age, sex, smoking status, and family history of PD. Competing risk models and inverse probability weighting were applied to address survivorship bias and missing data. Sex-stratified analyses explored potential differences by sex. Among 210,417 participants (mean follow-up = 9.17 years), 2592 developed PD. Higher exposure to both NO_2 and PM_{10} was associated with increased PD risk. In sex-specific models, the associations remained significant in males but not in females. Competing risk models confirmed elevated PD risk with NO_2 (HR = 1.05; 95 % CI: 1.01–1.09) and PM_{10} (HR = 1.08; 95 % CI: 1.03–1.13) in the overall cohort, with similar or stronger associations in males (NO_2: HR = 1.07; 95 % CI: 1.02–1.13; PM_{10}: HR = 1.07; 95 % CI: 1.02–1.14). In conclusion, long-term exposure to NO_2 and PM_{10} was linked to increased PD risk, particularly in males. These findings highlight the importance of incorporating sex-specific analyses in environmental research on PD.

1. Introduction

Parkinson's disease (PD) is one of the leading neurodegenerative disorders worldwide and poses an increasing burden as the global population ages (Collaborators, 2019). PD affects millions worldwide and has an estimated prevalence of 1–2 % in individuals over 60 years old. In the UK, approximately 145,000 individuals live with PD (UK, P.s., 2018). The pathology of PD is heterogeneous across the nervous system and no definitive cure is currently available (Kalia and Lang, 2015). Therefore, identifying risk factors for PD is crucial for effective early diagnosis and prevention.

Air pollution has been recognized as a risk factor for PD and plays an important role in its development (Murata et al., 2022; Hahad et al., 2020). Air pollutants include a variety of suspended particulate matter (PM) and gases, such as $PM_{2.5}$, nitrogen dioxide (NO_2), ozone (O_3), and carbon oxide (CO), all of which have been linked to an increased incidence of PD (Hahad et al., 2020; Han et al., 2020; Kasdagli et al., 2019). Possible mechanisms for air pollutants' link to PD include neuroinflammation, oxidative stress, and mitochondrial dysfunction. For example, exposure to particulate matter has been shown to induce inflammatory responses in the brain (Jaiswal and Singh, 2024). However, existing research and meta-analyses remain inconclusive in the associations (Han et al., 2020; Kasdagli et al., 2019; Hu et al., 2019), with some studies reporting non-significant or mixed results.

Nitrogen dioxide is one of the most prevalent air pollutants, primarily due to rising traffic and fuel consumption (EPA.gov, 2024). Three recent meta-analyses on air pollution and PD have identified positive associations between NO_2 exposure and PD risk, with aggregated relative risks (RR) of 1.03 (95 % CI: 0.99, 1.07), 1.01 (95 % CI: 0.98, 1.03), and 1.01 (95 % CI: 1.00, 1.03), respectively (Han et al., 2020; Kasdagli et al., 2019; Hu et al., 2019). However, two of three meta-analyses failed to achieve statistical significance. Among the reviewed studies, three

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reported non-significant associations (Liu et al., 2016; Chen et al., 2017a, 2017b), while the other three identified significant positive association (Finkelstein and Jerrett, 2007; Ritz et al., 2016; Shin et al., 2018). Interestingly, a large Italian cohort study found a possible negative association between NO_2 and PD development (Cerza et al., 2018a).

PM_{10}, another common air pollutant originating from industrial emissions, agricultural practices, traffic, and dust, has also been found to be controversially associated with PD. Two meta-analyses reported non-significant negative associations (Han et al., 2020; Kasdagli et al., 2019). A case-control study from Taiwan found that PM_{10} significantly increased the likelihood of developing PD (Liu et al., 2016). However, a separate study in the U.S. found an increased risk for PD only in women with no overall significant association (Chen et al., 2017a). In recent studies in Korean and Finnish populations, no significant association was observed between PM_{10} and PD (Jo et al., 2021; Rumrich et al., 2023). These inconclusive findings regarding both NO_2 and PM_{10} indicate the need for further research on their relationship with PD, particularly in representative large cohorts. In addition, although older adults are at highest risk for parkinsonian syndromes, these environmental exposures may also contribute to earlier onset, particularly in genetically susceptible individuals (Huang et al., 2024). To explore this possibility and potential subtype differences, we examined whether air pollution–parkinsonism associations extend to individuals < 60 years and vary across diagnostic groups (PD, PSP, MSA).

PD also presents differently based on sex: PD prevalence rate is 1.5–2 times higher and symptoms onset earlier in men compared to women (Haaxma et al., 2007). Understanding the sex differences in risk factors can inform tailored interventions. However, research on sex differences in PD risk related to air pollution exposure is scarce, and conclusive findings are yet to be established. Two Korean cohort studies indicated that air pollution may impact women's incidence of PD despite the greater prevalence of PD among men. Kim et al. showed that women had a higher risk for decreased cognitive function when exposed to PM_{10} and NO_2 while Lee et al. found an insignificant association, between NO_2 and PD exacerbation in women (Kim et al., 2019; Lee et al., 2017). Overall, the limited evidence available warrants further investigation into how air pollution impacts PD risk across sexes.

Air pollution has long been a significant health concern in the United Kingdom. In this study, we used the UK Biobank dataset to evaluate the associations between exposure to air pollutants PM_{10} and NO_2 and the onset of PD, as well as potential sex differences in these associations.

2. Methods

2.1. Study population

The data for this analysis was obtained from the UK Biobank database, a population-based cohort study involving over 500,000 individuals from across the United Kingdom. Participants, aged between 40 and 69, were recruited from 2006 to 2010. Detailed data collection procedures are available online (https://biobank.ndph.ox.ac.uk). Our study focused on a subset of individuals with air pollutant exposure data and a complete profile of selected covariates, including age, sex,

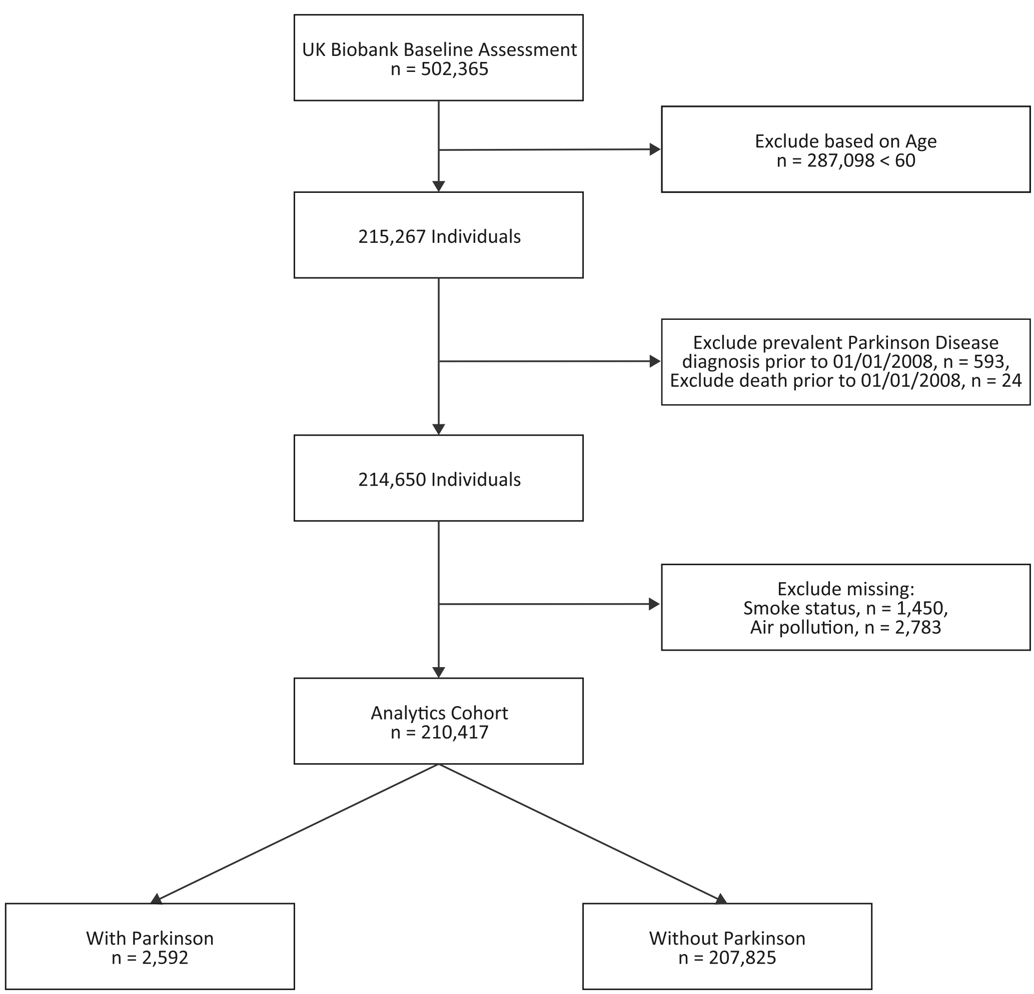

Fig. 1. Study flow chart for the complete case analysis of the analysis cohort.

smoking status, and family history of PD. In addition, we only included participants aged 60 years and older with no PD diagnosis or death record at the start of the follow-up, after the baseline air pollution exposure assessment (January 1st, 2008), to reduce potential confounding from early-onset, genetically driven Parkinson's disease. Exclusions due to any prevalent Parkinson diagnosis were 593 cases, while exclusions due to death records were 24 cases. An additional 4233 participants were excluded due to missing covariates and air pollutants data (Fig. 1). All participants provided written informed consent and ethical approval was granted by the North West Multi-centre Research Ethics Committee, the National Information Governance Board for Health & Social Care, and the Community Health Index Advisory Group.

2.2. Air pollutant measures

This study utilized estimates of annual exposure to NO_2 from 2005 to 2007 and PM_{10} in 2007. The UK Biobank acquired air pollution estimates from the Small Area Health Statistics Unit, as part of the BioSHaRE-EU Environmental Determinants of Health Project. Specifically, these data were derived from EU-wide air pollution maps with a resolution 100 m x 100 m). The location coordinates of UK Biobank participants were overlaid with these maps (projected to British National Grid) and were assigned the corresponding air pollution concentration from the relevant grid cell. The EU-wide air pollution maps were modelled based on a Land Use Regression (LUR) model for Europe which also incorporates satellite derived air pollution estimates to improve model performance (Vienneau et al., 2013). We focused on NO_2 from 2005 to 2007 and PM_{10} in 2007 because these pollutants had the most complete coverage and consistent modelling methods across years. Other annual air pollution estimates, including those for NO_2, NO, $PM_{2.5}$ and PM_{10} in 2010, though available in the UK Biobank database, were not included in the analysis due to the following reasons: 1) A different LUR model from the European Study of Cohorts for Air Pollution Effects was used to derive 2010 estimates and may introduce bias; 2) the 2010 estimates had substantial missingness (N = 41,296, 19.6 %), raising concerns about potential systematic bias and a lack of representativeness in the resulting cohort. Air pollutant exposures were scaled to continuous interquartile ranges (IQR) in the competing risk model.

2.3. PD diagnosis and time to PD onset or death

The diagnoses of PD were established using algorithms developed by the UK Biobank Outcome Adjudication Group (Bush et al., 2018). The algorithm was based on data from the UK Biobank baseline assessment data (verbal interview), linked hospital admissions records, and death registers. All-cause parkinsonism was identified, covering three subtypes: Parkinson's disease, progressive supranuclear palsy, and multiple system atrophy. The earliest record date was used as diagnosis date when multiple data sources for PD diagnosis were available. The algorithm's accuracy was validated in a study of 20,000 Biobank participants in Scotland. In our analyses, we excluded PD's diagnoses prior to January 1st, 2008, when the air pollution data were collected.

The time to PD or to death were calculated from the birth date to PD diagnosis or death date. Subjects who were lost to follow-up or did not have either event by the presumed last follow-up date (January, 2022) were considered as censored, and their time to censor was calculated from their birth date to January 1, 2022.

2.4. Covariates

Several demographic variables were examined, including age at baseline (in 2008), sex, ethnic background, education, smoking status, and family history of PD within the first generation. Age at baseline was calculated for the year 2008. Ethnic background was categorized as British and Non-British, while education was recoded as below or above college level. Smoking status was categorized as current, never, or previous smokers. Family history of PD was defined as 'Yes' when one of the study participant's biological parents had suffered from PD and 'No' otherwise. Specifically, income was categorized as follows: < £18,000, £ 18,000-£ 30,999, £ 31,000-£ 51,999, £ 52,000-£ 100,000, and > £100,000. Occupational exposure covariates related to the working environment (e.g., workplace very dusty, workplace full of chemicals or fumes, work with materials containing asbestos), categorized as do not know, rarely/never, sometimes, or often, were not included in the primary analysis due to high missingness (all approximately 78.8 %).

2.5. Statistical analysis methods

Summary statistics including means, medians, standard deviations (SDs), and interquartile ranges (IQRs) for continuous variables, frequencies and percentages for categorial variables, were examined. The percentages of missing data in each variable were also evaluated. The marginal association between each baseline demographic variable and PD incidence was assessed using Pearson's Chi-squared tests or Wilcoxon rank sum test, as appropriate. Variables with less than 20 % missing data that showed significant associations with PD were included as covariates in the subsequent regression models.

We assessed the associations between air pollutant exposures and PD incidence using logistic regression models adjusted for selected covariates. Socioeconomic variables such as household income (around 20.2 % missing) were excluded to avoid sample loss and selection bias, while education was excluded due to its minimal impact on the effect estimates. Separate models were developed for each air pollutant exposure variable. To account for potential survivorship bias, we conducted survival analyses with PD and death as two competing risks in Cox proportional hazard models, adjusted for the same set of the selected covariates. Hazard ratios (HR) and 95 % confidence intervals (CIs) of each pollutant for death and PD were reported. Both logistic and competing risk Cox PH models were refitted separately for male and female cohorts to understand the sex-specific associations. To control for multiple comparisons from the 4 exposure variables across years, we controlled overall False Discovery Rate (FDR) at 0.05 using the Benjamini-Hochberg method (Ferreira and Zwinderman, 2006). All statistical analyses were performed using R Version 4.3 (Chambers, 2008).

2.6. Sensitivity analysis

To assess potential differences between subjects included in the analysis and those excluded, we compared the demographic characteristics between the two groups using Pearson's Chi-squared tests and Wilcoxon rank sum test (Supplemental Table 1). If significant difference were observed, we conducted additional analysis using the inverse probability weighting (IPW) method (Wooldridge, 2007; Seaman and White, 2013; Horvitz and Thompson, 1952) under the assumption of missing at random (MAR). This probability was estimated using all demographic variables including age, sex, ethnic background, smoking status, PD family history, and BMI. Education and the deprivation index were also included to account for potential confounding by socioeconomic factors. The probabilities were then inversed and applied as weights in the regression models, thus giving more weights to observations that were less likely to be included. As sensitivity analyses for age restriction and diagnostic classification, we (i) repeated the primary models in a subgroup restricted to participants aged < 60 years and (ii) performed parallel subtype analyses for progressive supranuclear palsy (PSP) and multiple system atrophy (MSA); the same covariate set, and IPW procedure were used. A baseline characteristics table by diagnosis group (PD diagnosis, PSA, and MSA) was also included (Supplementary Table 3).

3. Results

3.1. Characteristics of participants

The final analysis cohort comprised 210,417 subjects, after exclusion of participants younger than 60 years old (N = 287,098), those with prevalent PD diagnosis (N = 593) or death records (N = 24) before the commencement of follow-up (January 1st, 2008), as well as those with missing data for all air pollution variables (N = 2783) or relevant covariates (N = 1450). Detailed characteristics of the study population are presented in Table 1. The mean age of included participants at the baseline year (2008) was 64.1 (SD = 2.89), with 110,928 (52.72 %) being female (Table 1).

A total of 2592 (12.3 %) subjects developed PD during the maximum 13.9 years of follow-up (mean (SD) = 9.17 (3.42) years). Compared to non-Parkinson subjects, those Parkinson cases were significantly older (65.16 versus 64.10 years), more likely to be male (62.92 % versus 47.09 %), less likely to be a current smoker (6.17 % versus 8.35 %), and more PD diagnosis in family history (6.60 % versus 4.01 %).

3.2. Associations of air pollutants with Parkinson incidence

In comparing included and excluded subjects, those included in our analysis were slightly younger, had a higher proportion of females, reported more family history of PD, were more likely to be of British origin, had higher education levels, and were less deprived (Supplemental Table 1). Therefore, we present the less biased results from the IPW method. Table 2 shows the associations between each air pollutant exposure and PD risk from the IPW adjusted logistic regression models.

In the overall sample, a trend towards increased odds of incident PD was observed for NO_2 across all years (2005: OR = 1.0044 per μg/m^3 increase, 95 %CI [1.0003, 1.0084], p = 0.029; 2006: OR = 1.0045 per μg/m^3 increase, 95 %CI [1.0001, 1.0089], p = 0.042; 2007: OR = 1.0045 per μg/m^3 increase, 95 %CI [1.0006, 1.0082], p = 0.019) and PM_{10} in 2007 (OR = 1.0229 per μg/m^3 increase, 95 %CI [1.0087, 1.0373], p = 0.001). In sex-specific analyses, exposures to NO_2 in 2005–2007 (2005: OR = 1.0062 per μg/m^3 increase, 95 %CI [1.0012, 1.0112], p = 0.014; 2006: OR = 1.0070 per μg/m^3 increase, 95 %CI [1.0014, 1.0125], p = 0.012; 2007: OR = 1.0063 per μg/m^3 increase, 95 %CI [1.0015, 1.0110], p = 0.008) and PM_{10} in 2007 (OR = 1.0232 per μg/m^3 increase, 95 %CI [1.0052, 1.0414], p = 0.013) were all significantly associated with an increased risk of PD in males, but not in females (2005: OR = 1.0012 per μg/m^3 increase, 95 %CI [0.9945, 1.0078], p = 0.717; 2006: OR = 1.0002 per μg/m^3 increase, 95 %CI [0.9928, 1.0074], p = 0.954; 2007: OR = 1.0013 per μg/m^3 increase, 95 %CI [0.9949, 1.0075], p = 0.676; PM_{10}: OR = 1.0223 per μg/m^3 increase, 95 %CI [0.9992, 1.0457], p = 0.056), although a positive trend was observed for PM_{10} in females.

3.3. Associations with time to PD onset

Table 3 and Fig. 2 show the association results between each air pollutant exposure and time to PD onset from the inverse-probability-weighted (IPW) cox competing risk models. The participants showed a 5 % increase in incident PD per IQR increment of NO_2 (2005: 95 %CI [1.01, 1.09]; 2006: 95 %CI [1.00 1.09]; 2007: 95 %CI [1.01, 1.09]), and an 8 % increase per IQR increment of PM_{10} (95 %CI [1.03, 1.13]). Similarly, all associations were significant solely in males, with no significant association in females. Specifically, in males, the HR for NO_2 exposure was 1.07 (95 %CI [1.02, 1.13]) in 2005, 1.07 (95 %CI [1.02, 1.13]) in 2006, and 1.07 (95 %CI [1.02, 1.12]) in 2007. The HR for PM_{10} exposure in 2007 was 1.07 (95 %CI [1.02, 1.14]) in males, but only a positive unsignificant sign in females (HR = 1.08, 95 %CI [1.01, 1.16]). The associations between air pollutant exposure and death as a competing risk are available in Supplemental Table 2. Results were also consistent in a sensitivity analysis restricted to participants aged < 60 years (Supplemental Table 4 & 5). In subtype analyses, PSP and MSA showed no statistically significant associations with NO_2 or PM_{10} exposure (Supplemental Table 6 & 7).

4. Discussion

In this study, we examined the association between PM_{10} and NO_2 exposure and PD development in 210,417 individuals from the UK Biobank cohort. We found significant positive associations between NO_2 or PM_{10} exposure and PD onset. These associations persisted in those < 60 years of age, suggesting effects beyond older adults, while no significant links were found for PSP or MSA, indicating possible subtype-specific patterns. We also provided novel evidence in sex differences in these associations, as these associations were significant in solely men. The observed sex disparities necessitate sex-sensitive approaches in PD research, as they may reflect biological or lifestyle differences that modulate vulnerability to environmental exposures.

Current literature has been inconclusive on the relationship between NO_2 exposure and PD development. While most cohort or case-control studies revealed positive associations, many studies failed to achieve statistical significance, with one study reporting a negative association (Cerza et al., 2018b). The varied results from existing studies were potentially due to different sample sizes, inclusion criteria, region of

Table 1
Demographics of included participants.

	Overall, n = 210,417		Male, n = 99,489		Female, n = 110,928	
Variable	Parkinson, n = 2592	Non-Parkinson, n = 207,825	Parkinson, n = 1631	Non-Parkinson, n = 97,858	Parkinson, n = 961	Non-Parkinson, n = 109,967
Age, mean (SD)	65.16 (2.90)	64.10 (2.89)	65.28 (2.90)	64.19 (2.90)	64.97 (2.89)	64.01 (2.88)
Sex, n (%)			-	–	-	-
Male	1631 (62.92)	97,858 (47.09)	-	-	-	-
Ethnicity, n (%)						
British	2366 (91.28)	190,902 (91.86)	1487 (91.17)	90,181 (92.15)	879 (91.47)	100,721 (91.59)
Education, n (%) College degree or equivalent	660 (25.46)	53,028 (25.52)	458 (28.08)	27,773 (28.38)	202 (21.02)	25,255 (22.97)
Smoke status, n (%)						
Never	1306 (50.39)	103,772 (49.93)	737 (45.19)	40,747 (41.64)	569 (59.21)	63,025 (57.31)
Previous	1126 (43.44)	86,705 (41.72)	796 (48.80)	42,277 (48.31)	330 (34.34)	39,428 (35.85)
Current	160 (6.17)	17,348 (8.35)	98 (6.01)	9834 (10.05)	62 (6.45)	7514 (6.84)
Family history, n (%)						
Parkinson within first generation	171 (6.17)	8343 (4.01)	111 (6.81)	3537 (3.61)	60 (6.24)	4806 (4.37)

Note: Baseline characteristics of sample size, categorized by sex (male, female). Nominal variables are shown as count (column percent), continuous variables are shown as mean (standard deviation).

Table 2
Adjusted association of air pollution exposure with incident PD.

	Parkinson, Median (IQR)	Non-Parkinson, Median (IQR)	Logistic IPW	
			Odds Ratio (95 % CI)	Raw P-value
Overall, n = 210,417				
NO2 2005	28.34 (22.73, 34.53)	27.76 (22.71, 33.71)	1.0044 (1.0003, 1.0084)	0.029*
NO2 2006	27.71 (22.39, 33.21)	27.33 (22.44, 32.47)	1.0045 (1.0001, 1.0089)	0.042*
NO2 2007	28.64 (23.31, 34.80)	28.32 (23.24, 34.10)	1.0045 (1.0006, 1.0082)	0.019*
PM10 2007	21.72 (20.28, 23.46)	21.64 (20.09, 23.34)	1.0229 (1.0087, 1.0373)	0.001**
Male, n = 99,489				
NO2 2005	28.40 (22.97, 34.65)	27.74 (22.66, 33.73)	1.0062 (1.0012, 1.0112)	0.014*
NO2 2006	27.88 (22.52, 33.41)	27.31 (22.39, 32.50)	1.0070 (1.0014, 1.0125)	0.012*
NO2 2007	28.84 (23.51, 34.94)	28.29 (23.18, 34.12)	1.0063 (1.0015, 1.0110)	0.008*
PM10 2007	21.70 (20.29, 23.36)	21.62 (20.08, 23.31)	1.0232 (1.0052, 1.0414)	0.013*
Female, n = 110,928				
NO2 2005	28.25 (22.36, 34.35)	27.78 (22.74, 33.69)	1.0012 (0.9945, 1.0078)	0.717
NO2 2006	27.51 (22.05, 32.88)	27.35 (22.47, 32.44)	1.0002 (0.9928, 1.0074)	0.954
NO2 2007	28.47 (23.08, 34.68)	28.34 (23.29, 34.09)	1.0013 (0.9949, 1.0075)	0.676
PM10 2007	21.77 (20.24, 23.60)	21.65 (20.10, 23.36)	1.0223 (0.9992, 1.0457)	0.056

Note: PD, Parkinson's Disease; IQR, inter-quartile range. Results were from logistic regression model. Model weighted using the inverse probability weighting (IPW) method. For overall PD sample adjusted for 4 covariates: age, sex, smoking status, and PD family history within first generation. For Parkinson sample stratified by sex, with 3 covariates: age, smoke status, and PD family history within first generation; Air pollution data unit: μg/m^3. Asterisk (*) indicates significance after FDR-adjustment. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

Table 3
IPW-weighted competing risk model association of air pollution exposure with time to PD.

	Overall		Male		Female	
	HR (95 % CI)	Raw P-value	HR (95 % CI)	Raw P-value	HR (95 % CI)	Raw P-value
NO2 2005	1.05 (1.01,1.09)	0.025*	1.07 (1.02, 1.13)	0.007*	1.01 (0.94, 1.08)	0.806
NO2 2006	1.05 (1.00, 1.09)	0.035*	1.07 (1.02, 1.13)	0.005*	1.00 (0.93, 1.07)	0.913
NO2 2007	1.05 (1.01, 1.09)	0.017*	1.07 (1.02, 1.12)	0.005*	1.01 (0.95, 1.08)	0.789
PM10 2007	1.08 (1.03, 1.13)	0.001**	1.07 (1.02, 1.14)	0.007*	1.08 (1.01, 1.16)	0.023

Note: Hazard ratios (HR) and 95 % confidence interval (CIs) for time to Parkinson's Disease (PD)'s diagnosis associated with IQR (interquartile range) of air pollution data, adjusted for sex, smoke status, and family history within first generation. Model weighted using the inverse probability weighting (IPW) method. Air pollutant exposures were scaled to continuous interquartile ranges (IQR). Asterisk (*) indicates significance after FDR-adjustment.
***$p < 0.001$.
* $p < 0.05$;
** $p < 0.01$;

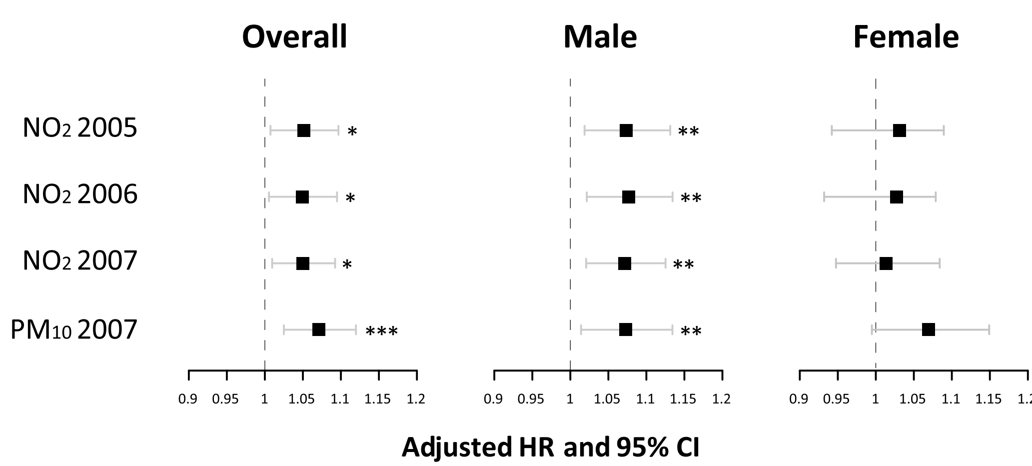

Fig. 2. Hazard ratio (HR) and 95 % confidence intervals (CIs) for incident PD associated with air pollution levels. Models are adjusted for sex, smoking status, and PD family history within first generation. Air pollution data unit: μg/m^3 . Asterisk (*) indicates significance after FDR-adjustment. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

interest, and adjusted covariates. A case-control study in Denmark found significant associations between NO_2 and PD when adjusted for age, sex, PD onset age, education, smoking history, family history of PD, and job history, but not when only age and sex were controlled (Ritz et al., 2016). A cohort study conducted in Ontario, Canada identified consistent positive association between aggregated NO_2 estimates up to 7 years before diagnosis and PD at various sensitivity analyses (Shin et al., 2018). However, in a population study in Rome, Italy from 2008 to 2013, a significant negative association was found between NO_2 and PD with age, birthplace, socioeconomic position, and marital status adjusted (Cerza et al., 2018b), while a case-control study in the US found no significant association between NO_2 and PD, adjusting for age, sex, race, education, caffeine intake, smoking status, and physical activity (Liu et al., 2016). A similar study conducted in Taiwan focusing on people at least 40 years old also failed to find significant association with age, sex, urbanization level, and comorbid disease adjusted (Chen et al., 2017a). Based on these studies, it is possible that adjusting for variables less relevant to PD onset and failing to adjust for highly relevant covariates such as smoking status and family history of PD may reduce the statistical significance in analysis. In addition, including middle-aged people younger than 60 years may also affect the association between PD and air pollution, given that early-onset PD between 40 and 60 years old may have a stronger genetic component compared to cases that occur later in life (Kolicheski et al., 2022).

At the same time, the average NO_2 level in studies showing significant positive associations, including the current work, were generally lower than 30 μg/m^3 (Finkelstein and Jerrett, 2007; Ritz et al., 2016; Shin et al., 2018), whereas the two studies with an average NO_2 level of 40–50 μg/m^3 reported non-significant or negative associations (Chen et al., 2017a; Finkelstein and Jerrett, 2007), indicating the level of air pollution may also affect the association between NO_2 and PD. A recent study analyzing the global trends in dementia and air pollution showed that NO_2 levels and dementia incidence were associated, though relatively weakly, but this association was stronger in the Global North countries (Guo et al., 2025). While this contrasts with previous cohort studies, which have vastly been conducted in Global North countries and have mixed results with the significance between NO_2 and PD associations, our study in a UK population, which is part of the Global North, supports Guo et al.'s findings. This discrepancy with previous cohort studies may arise from Guo et al.'s focus on Alzheimer's disease and related dementias over PD, and this study revealed broader, global trends over country-specific associations. Nevertheless, mitochondrial damage and oxidative stress have been shown to be mechanisms of neurodegeneration caused by NO_2 exposure, supporting the link between NO_2 and PD in our study (Li et al., 2022). Overall, our finding of significant positive association between NO_2 and PD agreed with most existing work and has added to the statistical strength of this relationship, highlighting the essential role of NO_2 in PD development. Our analysis was carefully tailored to participants at least 60 years old, with only covariates significantly associated with PD including age, sex, smoking status, and family history of PD adjusted. These methods may have contributed to the significant findings in the current work.

Similarly, the relationship between PM_{10} and PD incidence has yet to be fully clarified. The study in Rome, Italy reported a non-significant negative association between PM_{10} and PD, which turned significant in females (Cerza et al., 2018b). Similar results were also reported in studies in Taiwan, middle-aged females and males in the US, and Netherlands (Lee et al., 2016; Palacios et al., 2014, 2017; Toro et al., 2019). A US-based case-control study found a non-significant yet positive association between the two variables (Liu et al., 2016), and identified a sex difference where males showed a negative association, and the females showed a positive trend. In a Taiwan cohort, a significant positive association was found between PM_{10} and PD (Chen et al., 2017a). The mean PM_{10} level varied from 24.3 to 58.1 μg/m^3, with the PM_{10} level at different quartiles also differing. In addition, the trend of hazard ratio (HR) across quartiles was not consistent, indicating the relationship between PM_{10} and PD may not be linear. In three studies, the HR reached maximum when the PM_{10} level was around 20–30 μg/m^3 (Palacios et al., 2014, 2017; Toro et al., 2019; Gong et al., 2023). A recent global meta-analysis has associated PM_{10} with vascular dementia, albeit the association was not significant (Gong et al., 2023). While current conclusions on PM_{10} and its associations with PD and other dementias are still mixed, this may point more to geographical and economic differences, such as those differences suggested between the Global North and Global South (Guo et al., 2025). Additionally, across the literature, $PM_{2.5}$ generally has stronger, significant associations with dementia than PM_{10} (Pulmonology, 2025). This may be due to the smaller size of $PM_{2.5}$, enabling it to infiltrate deeper into the lungs via the lung-gas-blood barrier, and directly entering the brain through direct transmission across the olfactory nerve, bypassing the blood-brain barrier and damaging neurons and cerebral blood vessels (Li et al., 2022). In the current work, we found a significant positive association between PM_{10} and PD, with the median of PM_{10} being 22.6 μg/m^3, lower compared to most existing studies. The findings may have investigated the relationship between PM_{10} and PD at a different air pollution level that has not been covered by previous research. Future work investigating the PM_{10} -PD association at different PM_{10} ranges is warranted.

We found significant, positive associations between air pollutants and PD risk exclusively in men. Sex-stratified analyses have obtained mixed results. US-based cohort and case-control studies failed to identify significant associations in either males or females (Palacios et al., 2014, 2017). However, Liu et al. observed a trend of stronger associations in females for both NO_2 and PM_{10} (Liu et al., 2016). Ritz et al. identified significant association between NO_2 and PD in both sexes (Ritz et al., 2016). Salimi et al., despite not observing significance in the overall association, found stronger associations in males (Salimi et al., 2020). However, air pollution and genotype interactions may cause a greater risk of dementia development: Popov et al. observed that increased exposure to traffic air pollution worked synergistically with the APOE4 allele to decrease hippocampal volume in female, but not male, carriers of the allele in the UK Biobank (Popov et al., 2024). The presence of specific genetic markers could impact the extent of air pollution consequences, and this point could be further explored when differentiating sex-specific differences.

While exact relationship between air pollution and PD has not been fully clarified, there is evidence mechanistically to support significant associations in only men, as shown in the current work. Gaseous pollutants and particulates have been implicated to cross the blood-brain barrier and lead to neuroinflammation and oxidative stress in the frontal cortex, which may contribute to PD pathogenesis (Block and Calderón-Garcidueñas, 2009). In particular, PM_{10} and NO_2 have both been shown to have neurotoxic, inflammatory effects in the brain (Kim et al., 2024). However, females have been shown to be less susceptible to oxidative stress while men exhibit less efficient antioxidant mechanisms and higher basal inflammation (Martínez de Toda et al., 2023; Kander et al., 2017). Estrogen has been shown to exert neuroprotective effects through antioxidation, restraint of neuroinflammation, and modulation of dopaminergic pathways, potentially explaining resilience in females (Morale et al., 2006). Conversely, testosterone may exacerbate oxidative stress. Research has suggested a male-specific vulnerability to dopaminergic neurodegeneration through the expression of the SRY gene, which may increase dopaminergic neuron sensitivity to neurotoxicants (Adamson et al., 2022). In addition, men experience increased work-related exposures to air pollutants compared to women (Clougherty, 2010). While unequal gender distributions in the labor force may be a contributing factor, men report increased exposure to hazardous chemicals in mixed-gender occupations (Biswas et al., 2021). Therefore, higher levels of long-term exposure coupled with a comparatively weaker antioxidant mechanism may possibly provide the basis for the association between air pollution and PD in men.

Despite these insights, our study has limitations. The UK Biobank's findings may not be generalizable to other populations because its data is primarily drawn from White and British communities, where genetic and pollution level variability are limited. Because the cohort was restricted to adults ≥ 60 years, findings may also not generalize to younger-onset PD, which may involve different genetic and environmental mechanisms. Additionally, our focus on NO_2 and PM_{10}, rather than a broader range of pollutants, may limit the comprehensiveness of our conclusions. Although occupational exposures may partly explain the higher air pollution–parkinsonism risk observed in men, available workplace data was limited. Future research with detailed occupational exposure assessment is needed to clarify this potential confounding pathway. Finally, although we adjusted for education and area-level deprivation, incomplete income data limited our ability to capture all socioeconomic influences. Future research with more complete individual-level SES measures could help disentangle the effects of air pollution from social determinants of health.

In summary, our research supports the association between air pollution and PD development, emphasizing the need for sex-specific considerations in future studies. These findings may inform public policy aimed at mitigating environmental risks that contribute to PD.

CRediT authorship contribution statement

Rui Feng: Writing – review & editing, Supervision, Methodology, Conceptualization. **pan Michael:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis. **Haoer Shi:** Writing – review & editing, Writing – original draft, Methodology. **McKenna Sun:** Writing – review & editing, Writing – original draft. **Aimin Chen:** Writing – review & editing. **Jianghong Liu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Ethics approval and consent to participate

UK Biobank has received ethical approvals from the North West Multi-center Research Ethics Committee (MREC), the Community Health Index Advisory Group (CHIAG), the Patient Information Advisory Group (PIAG) and National Health Service National Research Ethics Service. Permission to use the UK Biobank resource for the research was approved by UK Biobank Access sub-committee (approved research application no. 88589).

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.neuro.2025.103353.

Data availability

The data that support the findings of this study are available on application to the UK Biobank.

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